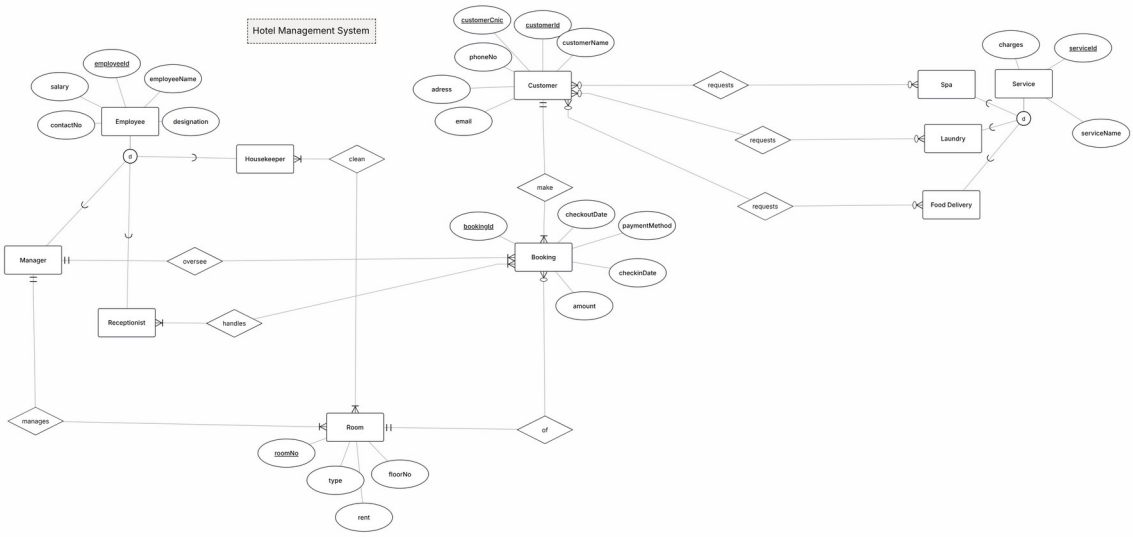
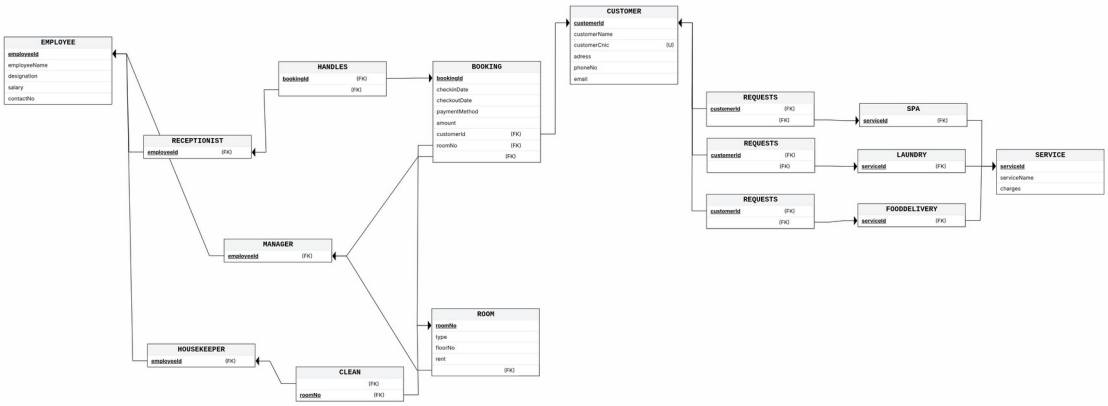
Practice Lab 04

ENTITY RELATIONSHIP DIAGRAM (ERD) And RELATIONAL SCHEMA

Q1. Business Case: Hotel Management System

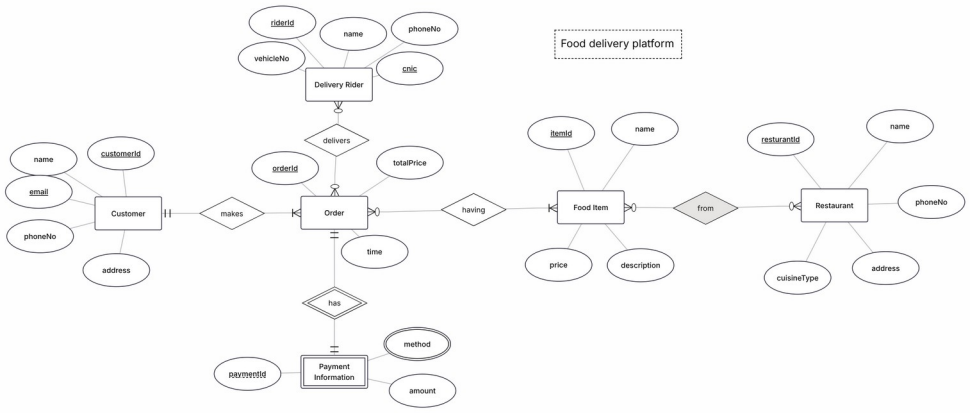
A luxury hotel in Karachi wants to maintain its operations digitally. The hotel has several rooms, and each room has a unique room number, type (Single, Double, Suite), floor number, and daily rent. Each room can be booked by one or more customers over time, but one booking can only be associated with one room. Customers have a unique customer ID, name, CNIC, address, phone number, and email address. A customer can make multiple bookings, and each booking includes check-in date, check-out date, payment method, and total amount. The hotel has several employees, including receptionists, housekeepers, and managers. Each employee has a unique employee ID, name, designation, salary, and contact number.
Each booking is handled by one receptionist, and one or more housekeepers are assigned to clean the booked rooms. The hotel also maintains a list of services such as laundry, food delivery, and spa, each with a unique service ID, name, and charges.
A customer may request one or more services during their stay, and each service request is linked to their booking.

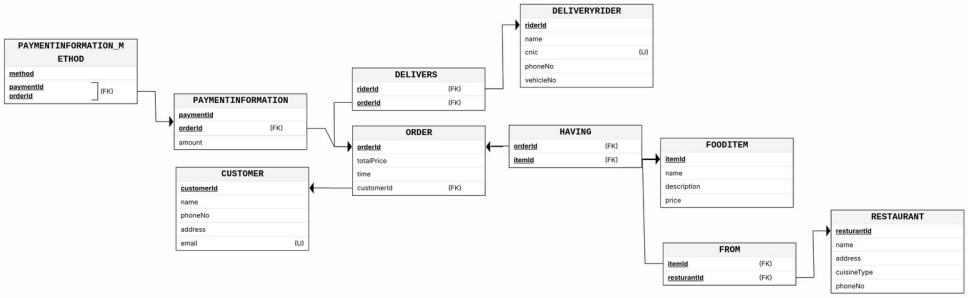



EMPLOYEE
employeeId
employeeName
designation
salary
contactNo
HANDLES
bookingId (FK)
(FK)
BOOKING
bookingId
checkinDate
checkoutDate
paymentMethod
amount
customerId (FK)
roomNo (FK)
(FK)
CUSTOMER
customerId
customerName
customerCnic (U)
adress
phoneNo
email
RECEPTIONIST
employeeId (FK)
REQUESTS
customerId (FK)
(FK)
SPA
serviceId (FK)
REQUESTS
customerId (FK)
(FK)
LAUNDRY
serviceId (FK)
SERVICE
serviceId
serviceName
charges
REQUESTS
customerId (FK)
(FK)
FOODDELIVERY
serviceId (FK)
MANAGER
employeeId (FK)
ROOM
roomNo
type
floorNo
rent
(FK)
HOUSEKEEPER
employeeId (FK)
CLEAN
(FK)
roomNo (FK)

Q2. Business Case: Online Food Delivery System

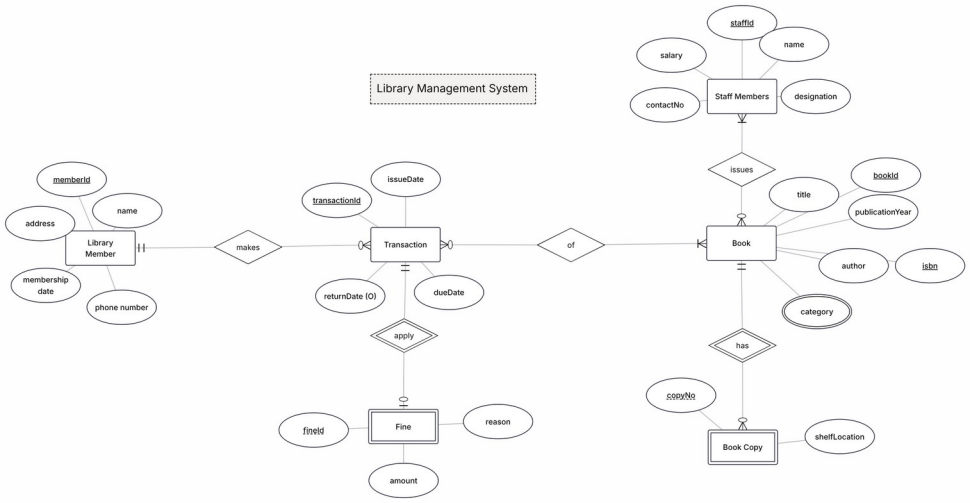
A food delivery platform called *QuickBite* operates in multiple cities. The system maintains data about restaurants, customers, orders, and delivery riders.
Each restaurant has a unique restaurant ID, name, address, cuisine type, and phone number. A restaurant offers multiple food items, and each food item has a unique item ID, name, description, and price.
A customer has a unique customer ID, name, phone number, address, and email. A customer can place multiple orders, and each order is identified by an order ID, order date, delivery status, and total amount.
Each order may contain one or more food items from a single restaurant. A delivery rider delivers the order. Each rider has a unique rider ID, name, CNIC, phone number, and vehicle number.
Payment information is also stored for each order, including payment ID, method (Cash, Card, Wallet), and amount paid.
A restaurant can receive many orders, but each order belongs to one restaurant only. Each rider can deliver multiple orders but only one at a time.

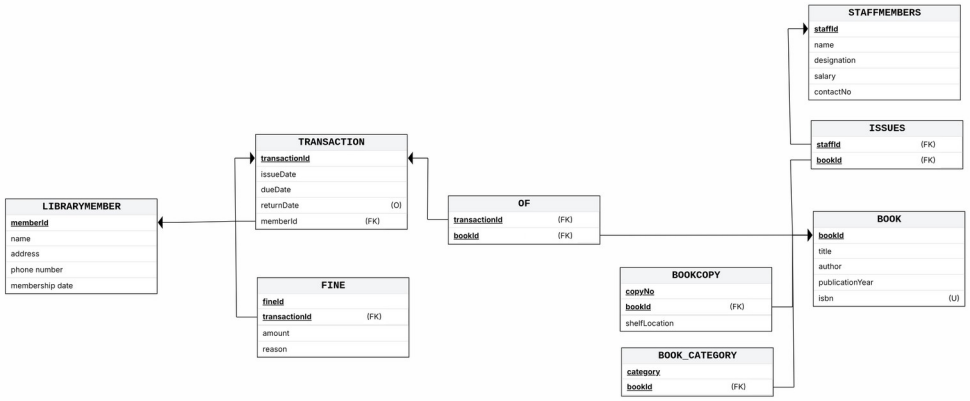



PAYMENTINFORMATION_METHOD
method
paymentId
orderId
(FK)
PAYMENTINFORMATION
paymentId
orderId (FK)
amount
DELIVERS
riderId (FK)
orderId (FK)
DELIVERYRIDER
riderId
name
cnic (U)
phoneNo
vehicleNo
ORDER
orderId
totalPrice
time
customerId (FK)
CUSTOMER
customerId
name
phoneNo
address
email (U)
HAVING
orderId (FK)
itemId (FK)
FOODITEM
itemId
name
description
price
FROM
itemId (FK)
resturantId (FK)
RESTAURANT
resturantId
name
address
cuisineType
phoneNo

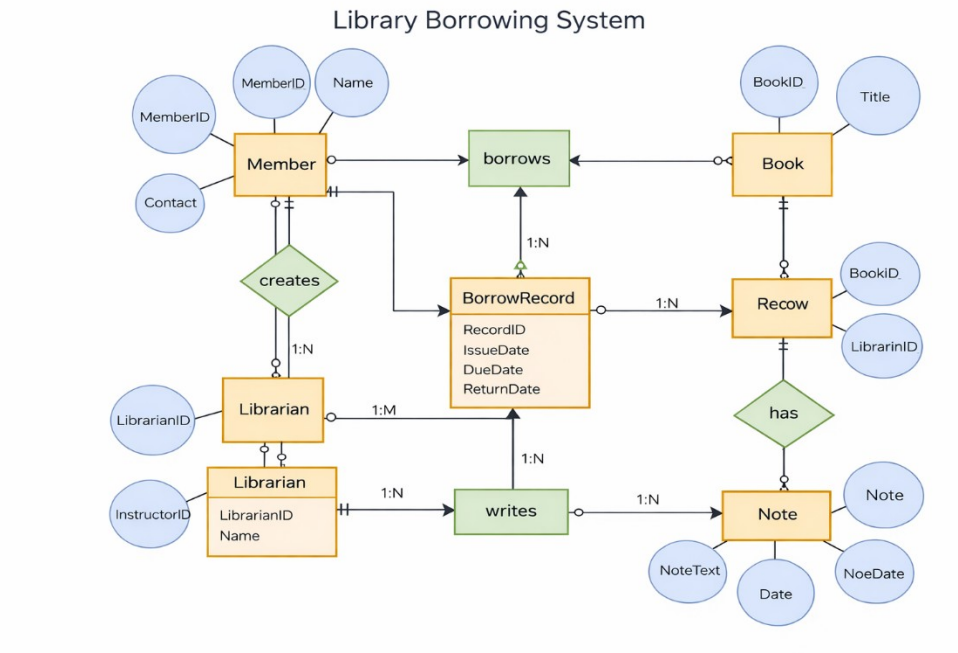
Q3. Business Case: Library Management System

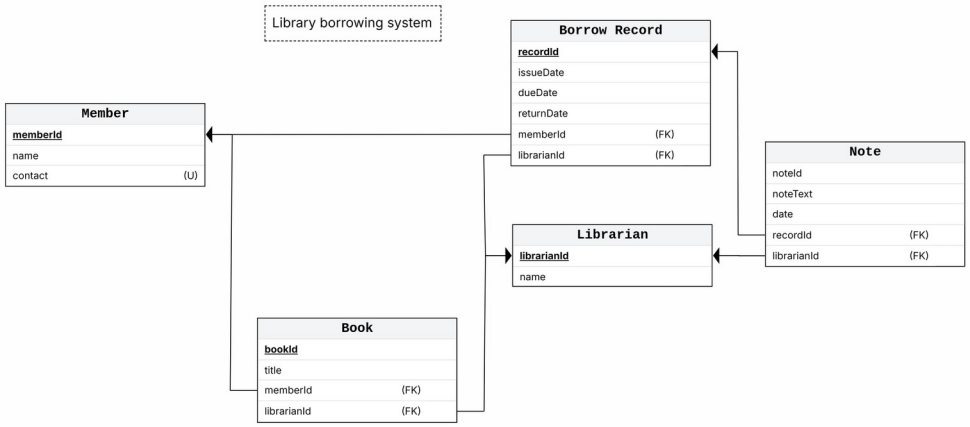
A public library in Islamabad maintains digital records for its members, books, and borrowing history.
Each book has a unique book ID, title, author, publication year, ISBN number, and category (Fiction, Science, History, etc.). A book can have multiple copies, each identified by a unique copy number and shelf location.
Library members have a unique member ID, name, address, phone number, and membership date. Members can borrow multiple books, but each book copy can be borrowed by only one member at a time.
Each borrowing transaction is recorded with a unique transaction ID, issue date, due date, and return date (if applicable).
The library has staff members, each with a unique staff ID, name, designation, salary, and contact number. A staff member issues and receives books for members.
The library also maintains information about fines for late returns, including fine ID, amount, and reason. Each fine is linked to a specific borrowing transaction.



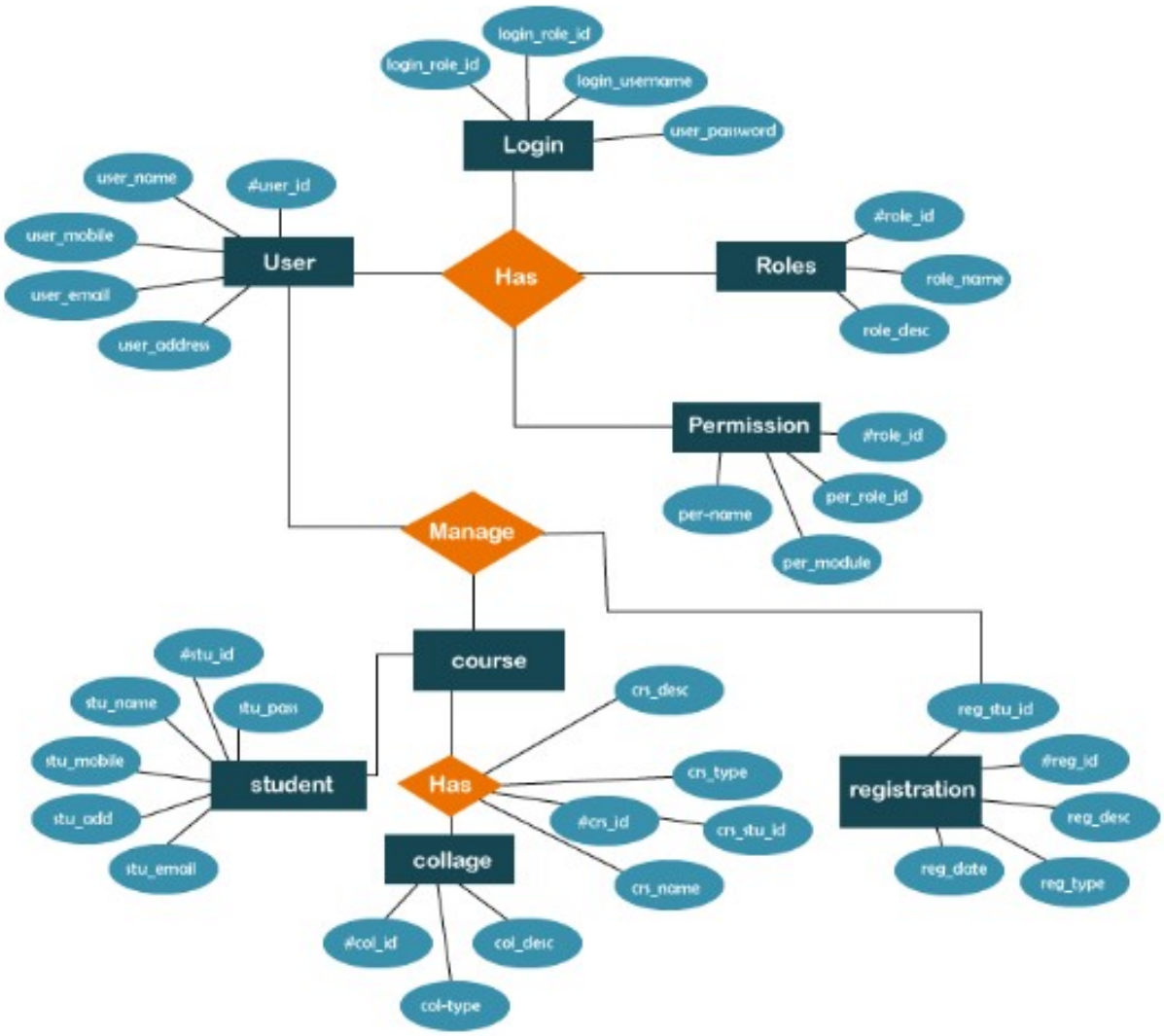

STAFFMEMBERS
staffId
name
designation
salary
contactNo
ISSUES
staffId (FK)
bookId (FK)
TRANSACTION
transactionId
issueDate
dueDate
returnDate (O)
memberId (FK)
LIBRARYMEMBER
memberId
name
address
phone number
membership date
OF
transactionId (FK)
bookId (FK)
BOOK
bookId
title
author
publicationYear
isbn (U)
FINE
fineId
transactionId (FK)
amount
reason
BOOKCOPY
copyNo
bookId (FK)
shelfLocation
BOOK_CATEGORY
category
bookId (FK)

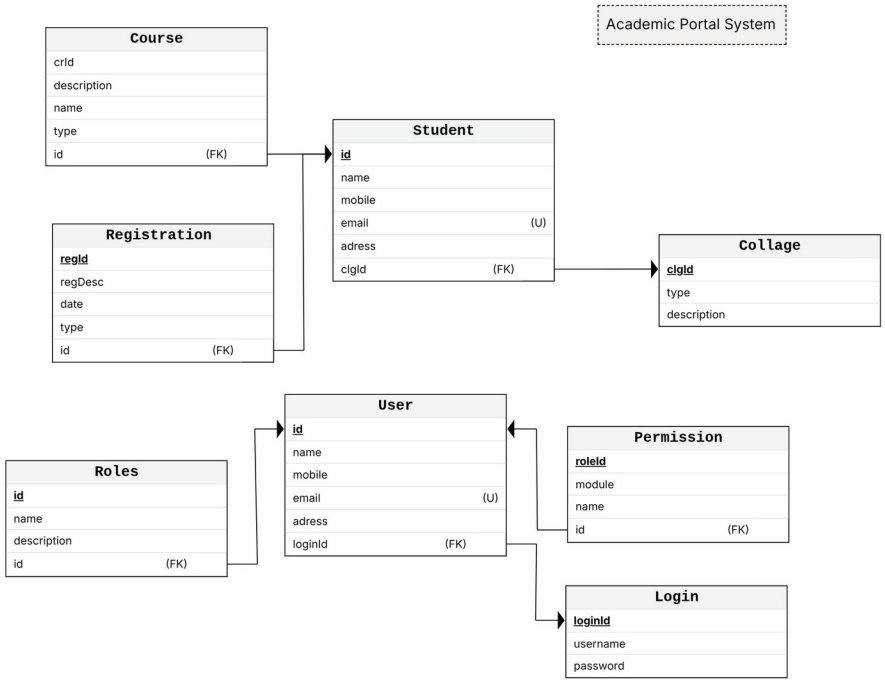
Q4.


Library Borrowing System
MemberID
Name
MemberID
Member
Contact
borrows
BookID
Title
Book
creates
1:N
BorrowRecord
RecordID
IssueDate
DueDate
ReturnDate
1:N
Recow
BookID
LibrarinID
1:N
LibrarianID
Librarian
1:M
has
1:N
Librarian
LibrarianID
Name
InstructorID
1:N
writes
1:N
Note
Note
NoteText
Date
NoeDate

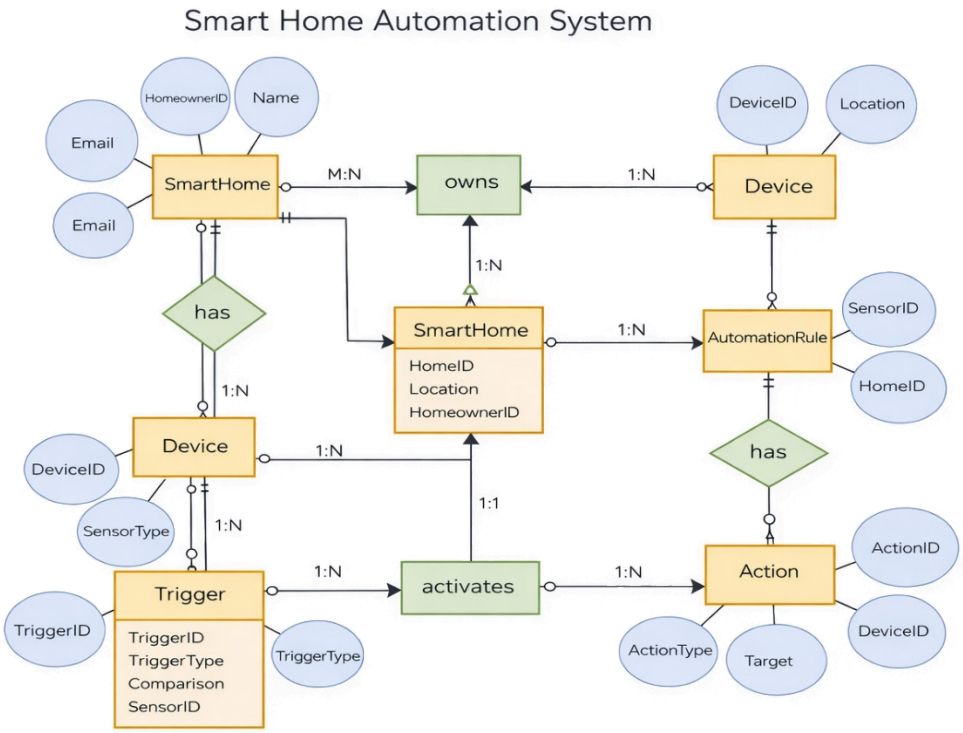

Library borrowing system
Borrow Record
recordId
issueDate
dueDate
returnDate
memberId (FK)
librarianId (FK)
Member
memberId
name
contact (U)
Note
noteId
noteText
date
recordId (FK)
librarianId (FK)
Librarian
librarianId
name
Book
bookId
title
memberId (FK)
librarianId (FK)

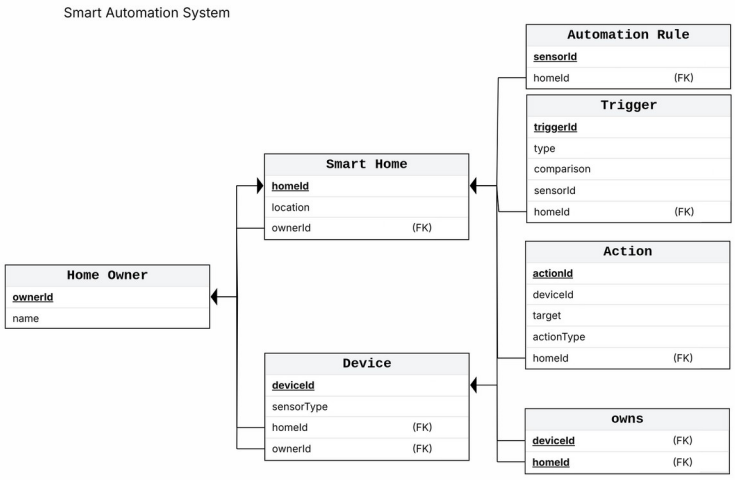
Q5.


login_role_id
login_role_id
login_username
user_password
Login
user_name
#user_id
user_mobile
User
Has
Roles
#role_id
role_name
user_email
role_desc
user_address
Permission
#role_id
per_role_id
per-name
per_module
Manage
course
#stu_id
stu_name
stu_pass
stu_mobile
student
stu_add
stu_email
Has
cns_desc
cns_type
#cns_id
cns_stu_id
cns_name
collage
#col_id
col_desc
col-type
reg_stu_id
#reg_id
registration
reg_desc
reg_date
reg_type


Academic Portal System
Course
crId
description
name
type
id (FK)
Student
id
name
mobile
email (U)
adress
clgId (FK)
Registration
regId
regDesc
date
type
id (FK)
Collage
clgId
type
description
User
id
name
mobile
email (U)
adress
loginId (FK)
Roles
id
name
description
id (FK)
Permission
roleId
module
name
id (FK)
Login
loginId
username
password

Q6:


Smart Home Automation System
HomeownerID
Name
Email
Email
SmartHome
M:N
owns
1:N
Device
DeviceID
Location
has
1:N
SmartHome
HomeID
Location
HomeownerID
1:N
AutomationRule
SensorID
HomeID
1:N
Device
DeviceID
SensorType
1:N
1:N
1:1
has
1:N
Trigger
TriggerID
TriggerID
TriggerType
Comparison
SensorID
TriggerType
1:N
activates
1:N
Action
ActionID
DeviceID
ActionType
Target


Smart Automation System
Automation Rule
sensorId
homeId (FK)
Trigger
triggerId
type
comparison
sensorId
homeId (FK)
Smart Home
homeId
location
ownerId (FK)
Action
actionId
deviceId
target
actionType
homeId (FK)
Home Owner
ownerId
name
Device
deviceId
sensorType
homeId (FK)
ownerId (FK)
Owns
deviceId (FK)
homeId (FK)